is sometimes called 'Zener voltage' and sudden increase in current is known as Zener current.

The breakdown or Zener voltage depends upon the amount of doping. If the diode is heavily doped, depletion larger will be thin and consequently the breakdown of the junction will occur at a lower reverse voltage. On the other hand, a lightly doped diode has a higher breakdown voltage. When an ordinary crystal diode is properly doped, so that it has a sharp breakdown voltage and it is called a Zener diode.

"A properly doped crystal diode which has a sharp breakdown voltage is known as a Zener diode"

10.5 TRANSISTOR

"When a third doped element is added to a crystal diode in such a way that two pn junction formed, the resulting device is known as a transistor

The transistor is a new type of electronic device and is capable of achieving amplification of weak signals.

There are two types of transistors, namely (i) n-p-n transistor (ii) p-n-p transistor.

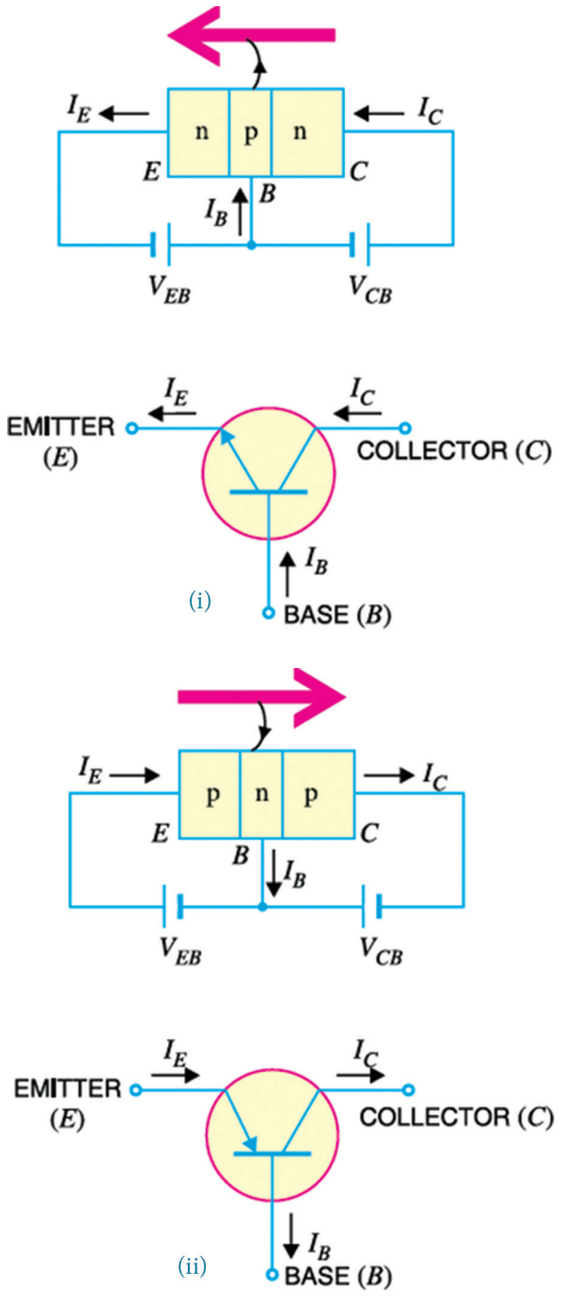


Fig 10.8 *Transistor Configuration*

The n-p-n transistor is composed of two n-type semiconductors seperated by a thin section of p-type as shown in fig 10.8(i). However a p-n-p transistor is formed by two p-sections seperated by a thin section of n-type as in fig 10.8 (ii).

A transistor (p-n-p or n-p-n) has three section of doped semiconductors.

EMA_English_Chp10.indd 160 12/17/2019 5:54:25 PM